

RED SEAL EXAM PREP

Gasfitter (Class A)

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Aligned to the published occupational standard topic weightings
<https://redsealquiz.ca> · Edition 2026-07

About This Material

Independent, unofficial study resource. This publication is not affiliated with, endorsed by, or approved by the Canadian Council of Directors of Apprenticeship (CCDA), Employment and Social Development Canada, the Red Seal Program, Skilled Trades Ontario, or any provincial or territorial apprenticeship authority. "Red Seal" is used solely to describe the examination for which this material provides practice.

Originality. All questions and explanations in this publication are original works created by the author. This material references only publicly available occupational standards (Red Seal Occupational Standard) for topic alignment and reproduces no content from any actual examination.

No guarantee of results. This study material is provided for educational and informational purposes only. Success on any certification examination depends on many factors, and no representation is made that using this material will result in passing any exam. Exam content, structure, and weightings are set by certification authorities and may change without notice.

Technical accuracy. Code values and technical specifications cited in explanations reflect commonly referenced Canadian standards at the time of writing. Always defer to the current edition of the applicable code or the manufacturer's documentation in real work.

Refunds. Because this is an instantly delivered digital file, all sales are final once the download link has been delivered. If you hit a technical problem — a corrupted or inaccessible file, or a duplicate purchase — contact us within 14 days and we will repair the file or refund you in full.

Updates. Corrections and additions are published to the same file. Re-download from your Ko-fi library to get the current edition at no extra cost.

© 2026 the author. All rights reserved. For personal study use by the purchaser only — redistribution, resale, or file sharing is prohibited.

Mock Exam — answer all questions, then check the key

Q1. What are the primary components of natural gas as distributed in Canada?

- A) Equal parts methane, propane, and butane
- B) Primarily methane (CH₄), typically 90-95%
- C) Primarily propane with methane added for odour
- D) Equal parts hydrogen and methane

Q2. A gas fitter calculates that the gas supply pressure at a building 300 m from the main drops to 4 in WC during peak demand. The appliances require a minimum inlet pressure of 5 in WC. What is the solution?

- A) Upsize the service pipe or request higher supply pressure from the utility
- B) Increase the appliance orifice size to compensate for the low pressure
- C) Accept 4 in WC — most appliances tolerate reduced supply pressure
- D) Install a booster regulator at the building to raise the delivered pressure

Q3. What are the early symptoms of carbon monoxide (CO) poisoning?

- A) Skin discolouration and high fever immediately upon exposure
- B) Strong smell similar to rotten eggs and burning sensation in throat
- C) Difficulty breathing due to CO being heavier than oxygen
- D) Flu-like headache, dizziness, nausea, and confusion

Q4. Why must every joint in a positive-pressure appliance vent be sealed and mechanically fastened?

- A) Because sealed joints let the vent be one size smaller than the tables
- B) Because the sealant is what carries the vent load between supports
- C) Because the flue gas is above room pressure, so a leak enters the room
- D) Because unsealed joints cool the vent below the flue gas dew point

Q5. What is second-stage regulation in a propane (LP gas) system?

- A) The second stage of pressure testing for LP systems
- B) The second regulator in a two-stage natural gas system at the meter
- C) The regulator reducing first-stage output to appliance pressure
- D) A backup regulator that activates if the primary fails

Q6. Approximately how many BTU of heat are in one cubic foot of natural gas?

- A) 1,000 BTU/ft³
- B) 2,500 BTU/ft³
- C) 500 BTU/ft³
- D) 5,000 BTU/ft³

Q7. A service regulator with a 60 psig inlet is set for 7 in WC outlet. With every appliance off, the lockup pressure at the test port reads 14 in WC. What decides whether this reading is acceptable?

- A) Whether it stays within the downstream piping and appliance limits
- B) Whether it stays within ten percent of the 7 in WC outlet setpoint
- C) Nothing — any lockup above the outlet setpoint condemns a regulator
- D) Nothing — lockup is only valid when read with an appliance firing

Q8. What information should a gas fitter record and provide to the building owner after commissioning a gas appliance?

- A) Only the installation date needs to be recorded
- B) Gas pressures, combustion readings and test results
- C) The appliance model and serial number, nothing more
- D) No documentation is required for residential gas work

Q9. A direct-vent (sealed combustion) gas appliance draws combustion air from where?

- A) From a combination of interior and exterior air
- B) From the furnace room through a draft hood
- C) From the exterior through a dedicated sealed air pipe
- D) From the building interior through a grille on the appliance

Q10. A mid-efficiency furnace and a natural draft water heater share a masonry chimney. The furnace is replaced by a condensing model vented in plastic pipe through the sidewall. What must be done about the water heater vent?

- A) Fit a barometric damper to replace the flue gas volume that was lost
- B) Nothing, because removing one appliance always improves the draft
- C) Re-size the chimney for the water heater alone and line it if oversized
- D) Connect the water heater to the new plastic sidewall vent and cap the chimney

Q11. When installing gas piping underground, what protection is required for metallic pipe?

- A) Approved corrosion coating and/or cathodic protection
- B) Paint with zinc-based coating and install in concrete conduit
- C) Install in corrugated plastic sleeve only
- D) No protection needed for schedule 40 steel pipe

Q12. At a winter service call the service regulator at a meter set is found buried in packed snow and ice, with its vent outlet covered. Why must that be cleared before the set is left in service?

- A) The vent is a rain drain, so a blockage causes rust in the spring case but nothing worse
- B) The vent is the breather for the meter index, so a blockage only stops it registering
- C) The vent must stay open for the regulator to control outlet pressure and relieve safely
- D) The vent supplies the air the regulator blends into the gas, so appliances will burn rich

Q13. A gas appliance rating plate states "Input: 100,000 BTU/h, AFUE: 80%". What is the useful heat output?

- A) 100,000 BTU/h — all input is usable heat
- B) 60,000 BTU/h — safety factor reduces output
- C) 80,000 BTU/h — 80% of input is converted to useful heat
- D) 120,000 BTU/h — efficiency multiplies the input

Q14. What is the difference between an intermittent pilot (IPI) and a direct spark ignition (DSI) system?

- A) IPI is for high-efficiency furnaces only; DSI is for boilers only
- B) They are identical ignition systems with different names
- C) IPI uses a standing pilot; DSI uses a hot surface ignitor
- D) IPI sparks a pilot first; DSI sparks the main burner directly

Q15. A condensing boiler is installed with a return water temperature of 38°C (100°F). What is the implication of this return temperature?

- A) It enables condensing operation, achieving the highest efficiency
- B) Return temperature is too low — the boiler will short-cycle
- C) Low return temperature causes carbonic acid corrosion of the heat exchanger
- D) The boiler is inefficient — return temperature must be above 60°C

Q16. What are the four categories of gas appliance venting classification?

- A) Grades I-IV based on appliance efficiency
- B) Classes 1-4 based on appliance BTU rating
- C) Types A, B, C, D based on flue pipe material
- D) Categories I-IV, by vent pressure and flue temperature

Q17. A gas furnace is to be installed at a site about 1,000 m above sea level. What determines how the appliance is set up for that elevation?

- A) Nothing extra, because the appliance regulator compensates for thin air
- B) The gas utility, which lowers the pressure it delivers to suit the elevation
- C) A larger burner orifice, because the thinner air lets it accept more gas
- D) Its high-altitude certification and the manufacturer's instructions

Q18. On a large commercial gas boiler, the burner management system runs the combustion air fan for a timed period before the ignition trial is allowed to begin. What sets the required length of that pre-purge?

- A) The delay the flame safeguard needs before it can prove the pilot
- B) Chamber and flue volume, and the number of air changes required
- C) The time the refractory needs to warm up and avoid thermal shock
- D) The warm-up time the hot surface ignitor needs

Q19. What is the most common pipe material for above-ground natural gas distribution within buildings?

- A) Type L copper tubing
- B) Schedule 40 PVC plastic pipe
- C) Galvanized steel pipe
- D) Black steel pipe

Q20. What is the purpose of an inducer (induced draft) motor on a mid-efficiency or high-efficiency gas furnace?

- A) To supply combustion air to the burner before ignition
- B) To pull combustion gases through the heat exchanger to the flue
- C) To induce a draft in the chimney by heating the flue pipe
- D) To circulate conditioned air through the building duct system

Q21. A gas-fired hot water boiler will not fire, and the fault is traced to its low-water cut-off. What condition is that control there to protect against?

- A) Water falling below safe level, which would let the boiler dry-fire
- B) Boiler pressure climbing above the relief valve setting on the vessel
- C) The circulating pump losing prime and running dry
- D) Return water temperature dropping far enough to condense flue gas

Q22. What governs the maximum spacing between supports on a horizontal run of rigid black steel gas pipe?

- A) A fixed 3 m (10 ft) for every size and material of pipe
- B) The pipe size, read from the support-spacing table in the code
- C) A fixed 1 m (3 ft) for every size and material of pipe
- D) Nothing beyond a support at each threaded fitting on the run

Q23. Where should a combustible gas detector be positioned for a natural gas leak vs. a propane leak?

- A) Always at floor level for both natural gas and propane
- B) Always at breathing height (1.2 m) for both gases
- C) Natural gas: near the ceiling; propane: near the floor
- D) Detector position does not matter — both gases disperse evenly

Q24. What must a Gasfitter Class A gas fitter verify about vent termination height above a flat roof?

- A) Vent must extend to the highest point of the building
- B) Vent can terminate at roof level — no height requirement
- C) 150 mm above the roof is sufficient for all appliances
- D) A minimum height above the roof surface per CSA B149.1

Q25. A gas fitter is one coupling short on a certified plastic vent system and has a same-size fitting, primer and cement from a different manufacturer's vent kit on the van. What is the correct decision?

- A) Use them, provided the substituted joint is wrapped in foil tape once it has cured
- B) Use them, since same-size plastic vent parts are interchangeable between makers
- C) Use them, provided the finished vent passes a tightness test at the terminal after
- D) Do not mix them — a certified vent is pipe, fittings, primer and cement together

Q26. A natural gas piping system supplied at 14 in WC or less is being sized with the capacity tables in Annex A of the installation code. Which length is used to enter the table?

- A) The total of the measured lengths of every section in the system, since all of it adds to the pressure drop
- B) The measured length of the run from the meter or last-stage regulator to the most remote outlet
- C) The measured length of each section on its own, so that a short branch is sized on its own short length
- D) The measured length of the run to the most remote outlet plus an equivalent length for every fitting

Q27. A Gasfitter Class A gas fitter is commissioning a new gas boiler. What is the correct sequence for commissioning?

- A) Verify supply → purge piping → light → verify controls → document
- B) Light appliance → check gas pressure → test safety controls → check combustion
- C) Check combustion first, then adjust controls → verify gas supply
- D) Test safety controls → light appliance → purge piping → check pressure

Q28. What thread sealant must be used on gas pipe NPT connections?

- A) Standard white PTFE tape (plumber's tape)
- B) No sealant needed for tapered NPT threads — they self-seal
- C) Yellow PTFE tape or gas-rated pipe thread compound
- D) Loctite thread sealant (any grade)

Q29. What is the difference between a service regulator and an appliance regulator?

- A) Service regulators are indoors; appliance regulators are outdoors
- B) They are interchangeable, since both do the same job in a gas system
- C) Appliance regulators handle higher pressures than service regulators
- D) Service reduces distribution pressure; appliance reduces it further

Q30. How do the Red Seal trade descriptions distinguish a Gasfitter Class A from a Gasfitter Class B?

- A) Class A is a regulated trade; Class B is unregulated across Canada
- B) Class A may work above 400,000 Btuh (120 kW); Class B may not
- C) Class A is the apprentice level; Class B is the journeyman level
- D) Class A covers natural gas only; Class B covers propane only

Q31. What does the colour of a gas burner flame indicate about combustion quality?

- A) Blue = complete combustion; yellow = incomplete combustion
- B) Blue flame = too much air; yellow flame = correct mixture
- C) Flame colour is decorative and has no combustion significance
- D) Yellow is normal for natural gas; blue is normal for propane only

Q32. During a combustion analysis, the CO₂ reading in the flue gas of a natural gas furnace is 8.5%. What does this indicate?

- A) Excess air is diluting the CO₂ below its stoichiometric maximum
- B) 8.5% CO₂ indicates perfect combustion efficiency
- C) CO₂ has no relevance to combustion quality analysis
- D) 8.5% CO₂ is too high — CO₂ should never exceed 5% in flue gas

Q33. Tubing is being run inside a hollow stud wall. Under CSA B149.1, what determines whether it must be protected at the studs, joists and plates with No. 16 USG (1.59 mm) steel plates or sleeves?

- A) Whether it lies within 1.75 in (43 mm) of the wall surface
- B) Whether the operating pressure in it is above 14 in WC (3.5 kPa)
- C) Whether the run inside the cavity is longer than 6 ft (2 m)
- D) Whether it is CSST; copper tube is protected by its own wall

Q34. To confirm that a furnace is firing at its rated input, a gas fitter clocks the gas meter with every other gas appliance shut off. What does that test measure directly?

- A) The volume of gas the appliance draws over a measured length of running time
- B) The manifold pressure that the appliance regulator is holding at the burner
- C) The efficiency of the appliance, once the flue gas temperature is deducted
- D) The heating value of the gas the utility is delivering to that meter set

Q35. What is the Lower Explosive Limit (LEL) of natural gas and what does it mean?

- A) 0.5% — the threshold for immediate evacuation
- B) 15% natural gas concentration — the maximum explosive level
- C) 1% natural gas concentration — below this level gas will not ignite
- D) 5% gas in air — the minimum concentration that can ignite

Q36. What happens to gas pressure in a pipe as flow rate increases and why does this matter for pipe sizing?

- A) Pressure increases with flow — pipe must be sized to withstand higher pressure
- B) Pressure drops from friction — pipes are sized to limit the drop
- C) Pressure is constant regardless of flow rate in a properly sized system
- D) Pressure increases then decreases — only the midpoint matters for sizing

Q37. What is the standard low-pressure natural gas outlet pressure from a service regulator to a residential building?

- A) 7 in WC (1.75 kPa)
- B) 14 in WC (3.5 kPa)
- C) 2 psig (13.8 kPa)
- D) 3.5 in WC (0.87 kPa)

Q38. What is the maximum horizontal run permitted for a power-vent (Category III) appliance vent?

- A) No limit — power venting can run any distance horizontally
- B) As specified by the appliance manufacturer
- C) 600 mm (2 ft) — power vents must be nearly vertical
- D) Maximum 3 m regardless of appliance size

Q39. A gas meter is rated for a maximum capacity of 250 ft³/h. The total connected load for the building is 400,000 BTU/h of natural gas appliances. Is the meter adequate?

- A) Yes — 250 ft³/h is always sufficient for residential use
- B) Yes — connected load does not equal actual demand
- C) Cannot determine without knowing the number of appliances
- D) No — the required 400 ft³/h exceeds the meter capacity

Q40. At what flue gas temperature will condensation occur in a natural gas appliance vent, and why does this matter?

- A) There is no specific dew point — condensation depends only on ambient temperature
- B) Condensation occurs at 100°C (212°F) — the boiling point of water
- C) About 55-60°C (130-140°F) — below this, acidic condensate forms
- D) Condensation only occurs in propane appliances — natural gas flue gas does not condense

Q41. What is the correct procedure for purging a new gas piping system before lighting appliances?

- A) Light the pilot repeatedly — purging happens automatically during ignition attempts
- B) Open all appliance valves and let gas flow for 60 seconds to purge the system
- C) Purge using nitrogen, then switch to gas
- D) Slowly displace air with gas, venting to a safe outdoor location

Q42. What is stoichiometric combustion?

- A) Combustion at the highest possible flame temperature
- B) Combustion with insufficient air, producing maximum heat output
- C) Combustion with exactly the theoretical air required
- D) Combustion with excess air — the safest type for gas appliances

Q43. A high-efficiency condensing furnace produces liquid condensate that drains from the heat exchanger. What is special about this condensate?

- A) It is slightly acidic and may require a neutralizer
- B) It contains natural gas byproducts and is considered hazardous waste
- C) It is alkaline and must be diluted before discharge
- D) It is pure water that can be discharged anywhere

Q44. What is a hot surface ignitor (HSI) and how does it work in a modern gas furnace?

- A) A UV flame sensor that monitors the main burner flame
- B) A spark ignitor that creates a high-voltage arc to light the burner
- C) A pilot burner that remains lit continuously between heating cycles
- D) An element that glows red-hot to ignite the gas burner on contact

Q45. What is the purpose of a drip leg (sediment trap) installed before a gas appliance?

- A) To trap condensate, debris, and scale from the gas supply
- B) To serve as an emergency shutoff in case of fire
- C) To reduce gas pressure to the appliance
- D) To allow pressure testing of the appliance separately

Q46. What is a lockout/tagout (LOTO) procedure for gas systems and when is it required?

- A) LOTO is for electrical systems only — gas systems use manual shutoff valves
- B) LOTO for gas means attaching a warning label to the appliance
- C) Locking and tagging the closed gas valve before maintenance
- D) Only required for high-pressure industrial gas systems

Q47. When two gas appliances share a common vent, what must be considered?

- A) The larger appliance vents normally; the smaller one must use a separate vent
- B) Only matching appliance types (e.g., two furnaces) may share a common vent
- C) The common vent must be sized for the combined flue gas volume
- D) Common venting always reduces vent size requirements

Q48. What is the purpose of the listed cap fitted at the top of a Type B vent?

- A) It filters carbon monoxide out of the flue gas before it leaves
- B) It restricts the outlet so flue gas leaves faster and draft rises
- C) It cools the flue gas so the terminal is safe to touch when hot
- D) It keeps rain out and limits wind-induced downdraft at the terminal

Q49. What is flame rectification (flame sensing) and how does it work in modern gas furnaces?

- A) A method to correct the shape of an abnormal flame using air baffles
- B) A safety circuit where a thermocouple generates DC voltage to prove pilot flame
- C) The flame conducts DC current, proving the burner is lit
- D) A sensor that detects the blue colour of a normal gas flame

Q50. A gas dryer venting system must terminate where?

- A) Into a laundry room if it has a window
- B) Into an approved lint trap inside the building
- C) Into the attic space to reduce moisture in the house
- D) To the exterior of the building only

Answer Key & Explanations

Q1. Answer: B

Pipeline natural gas is primarily methane (CH₄), typically 90-95%, with minor amounts of ethane (C₂H₆), propane (C₃H₈), and butane (C₄H₁₀), plus inert gases (N₂, CO₂). The exact composition varies by source and region. This composition determines energy content (~1,000 BTU/ft³) and combustion characteristics.

Q2. Answer: A

The appliances need at least 5 in WC at their inlets and the system delivers 4 in WC at peak, so either too much pressure is being lost in the run or the supplied pressure is too low to begin with. Both are corrected on the supply side: increase the service pipe diameter so the drop over the 300 m falls, or arrange with the utility for a higher delivered pressure and regulate down at the building. A regulator adds no energy to the gas — it can only reduce and control a pressure that is already there — so a booster regulator has nothing to boost when the upstream pressure itself is inadequate. Opening up the orifices is worse than useless: orifice size is matched to the appliance and to its manifold pressure, and enlarging it to chase a low supply produces an over-fired, badly burning appliance outside its certification. Living with 4 in WC is not an option either, because an appliance below its rated inlet pressure under-fires, burns poorly and can produce CO.

Q3. Answer: D

CO (carbon monoxide) is odourless and colourless. Early symptoms: headache, dizziness, nausea, weakness, and confusion — often mistaken for the flu. Key distinguishing feature: symptoms improve when leaving the building (away from CO source) and worsen upon return. Severe exposure causes unconsciousness and death.

Q4. Answer: C

In a Category III or Category IV system the appliance fan pushes the flue gas, so the inside of the vent sits above the pressure of the room along its whole length. Any gap at a joint therefore pushes combustion products, carbon monoxide included, out into the building. A natural draft vent behaves the opposite way: it runs below room pressure, so a loose joint draws room air inward and dilutes the flue gas instead of releasing it. That difference is why positive-pressure vents are assembled with the listed gaskets, cements or solvent-welded joints plus the mechanical fasteners the vent system calls for, and why the joints are checked for leakage at start-up. Sealing does not change the vent size the sizing tables require, and it is never a substitute for proper support.

Q5. Answer: C

LP gas systems typically use two-stage regulation: First stage at the tank reduces vapor pressure (which varies with temperature) to a constant ~10 psig (69 kPa). Second stage at or near the building reduces 10 psig to appliance operating pressure (~11 in WC / 2.74 kPa). This provides stable pressure regardless of tank level or temperature.

Q6. Answer: A

Natural gas has an energy content of approximately 1,000 BTU per cubic foot (higher heating value). This is a fundamental value for pipe sizing, appliance sizing, and combustion calculations. Propane contains approximately 2,500 BTU per cubic foot (as gas).

Q7. Answer: A

Lockup is the outlet pressure a regulator settles at once downstream demand falls to zero, so it is read with the appliances off, not while one is firing. Some rise above the setpoint is normal, because the seat needs overpressure to close fully, and the measured value on its own therefore neither condemns nor clears the regulator. Acceptability is a comparison: the measured lockup must stay at or below the design pressure of the downstream piping, at or below the maximum inlet pressure marked on the rating plate of every appliance downstream, and within the lockup limit published for that regulator. If it exceeds any one of those, the regulator is readjusted or replaced before the system is left in service. There is no general percentage-of-setpoint rule to fall back on, so take the allowable values from the current CSA B149.1 requirements and the equipment data for the system in front of you.

Q8. Answer: B

Professional commissioning documentation provides proof of safe installation and a baseline for future service. Records should include: gas supply (inlet) pressure measured at the appliance, manifold (burner) pressure downstream of the appliance regulator, combustion analysis readings (CO₂, O₂, CO, flue temperature, efficiency), safety control verification results, vent check, and any deficiencies noted or corrected. Some jurisdictions require documentation to be submitted to the authority as well as given to the owner.

Q9. Answer: C

Direct-vent (sealed combustion) appliances use a two-pipe system: one pipe draws combustion air from outside, and the other exhausts flue gases to outside. The combustion process is completely isolated from the building interior, making them ideal for tight buildings and eliminating combustion air requirements.

Q10. Answer: C

This is the orphaned appliance problem. The chimney was sized for the combined input of two appliances, and the water heater is now the only load on it. Its much smaller volume of flue gas cannot keep that flue warm, so the gases cool on the way up, draft weakens and moisture condenses inside the chimney, which leads to spillage at the draft hood and long-term damage to the masonry. The chimney must be re-evaluated against the vent sizing tables for the water heater input alone, and in most cases a correctly sized listed liner is installed. Connecting the water heater to the new appliance vent is not an option, because that vent is a sealed positive-pressure system and the plastic is not rated for natural draft water heater flue gas temperature. A barometric damper adds dilution air, which would cool the flue further rather than help. Confirm the combustion air supply and verify draft at the water heater before leaving the job.

Q11. Answer: A

Buried metallic gas pipe is attacked by soil moisture and by stray currents, so it has to be protected against corrosion before the trench is closed. The accepted methods are a factory-applied polyethylene coating or wrap, a field-applied tape wrap, or a fusion-bonded epoxy coating. Cathodic protection is a supplementary method, added where the soil conditions, the system or the authority having jurisdiction call for it; it is not an automatic requirement on every buried steel line, and treating it as one is a common misreading. The coating has to be carried over fittings and joints as well as over the pipe barrel, and any damage done in handling or backfill must be made good before the trench is closed, because a break in the coating concentrates the corrosion at that one spot instead of spreading it. Bare schedule 40 steel in the ground is exactly what fails. A zinc-rich paint is a shop finish, not a buried-pipe coating system, and setting the pipe in concrete conduit traps moisture against it rather than keeping moisture off it. A loose corrugated plastic sleeve gives mechanical protection only: it is not a corrosion coating, and it can hold water against the pipe. Confirm the accepted coating systems and any cathodic-protection requirement against the edition of CSA B149.1 in force in your jurisdiction and the provincial amendments.

Q12. Answer: C

A diaphragm regulator works by balancing gas pressure under the diaphragm against a spring, with the far side of the diaphragm open to atmosphere through the vent. Block that vent and the regulator loses its atmospheric reference and can no longer hold the outlet where it was set. The vent is also the relief path: if the diaphragm splits or the seat fails to close, gas is meant to discharge from the vent to a safe location outdoors. CSA B149.1's definition of a safe location for venting gas, reproduced in Alberta Municipal Affairs' published comparison of the 2020 edition, is one where you can reasonably expect to prevent, among other things, 'blocking of the vent termination by snow, ice, water, or any other object or thing' — so a vent packed with ice is by definition not in a safe condition. TSSA and Enbridge Gas make the same point to homeowners: a gas meter covered in snow or ice can impede its function of controlling the pressure of the gas supplied to a home. Clear it gently with a soft brush, never by kicking, hitting or chipping at it. The vent is not a rain drain, it does not feed air into the gas stream, and it has nothing to do with the meter index.

Q13. Answer: C

AFUE of 80% means 80% of fuel energy is converted to heat delivered to the space. $\text{Output} = \text{Input} \times \text{AFUE} = 100,000 \times 0.80 = 80,000 \text{ BTU/h}$. The remaining 20,000 BTU/h (20%) is lost through the flue.

Q14. Answer: D

IPI (Intermittent Pilot Ignition): a spark creates a pilot flame at the start of each heat cycle; the flame sensor confirms pilot is lit, then the main gas valve opens and the pilot ignites the main burner. DSI (Direct Spark Ignition): the spark fires directly at the main burner without a pilot. DSI is faster but requires the spark to be at the burner.

Q15. Answer: A

Condensing boilers are designed to operate at low return water temperatures (below 55°C/130°F). At low temperatures, the flue gases cool below their dew point and condense, releasing latent heat. This is the condensing effect — extracting extra energy from the fuel, achieving the highest efficiency (90%+ AFUE). High return temps prevent condensing and reduce efficiency to non-condensing levels.

Q16. Answer: D

Appliances are classed Categories I-IV based on vent static pressure (positive or non-positive) and flue gas temperature relative to dew point. Per CSA B149.1: Cat I = non-positive pressure, above dew point (80% furnace). Cat II = non-positive pressure, below dew point (condensing, low temp). Cat III = positive pressure, above dew point (some direct vent). Cat IV = positive pressure, below dew point (condensing, high efficiency). Venting material requirements differ for each.

Q17. Answer: D

Air thins with elevation, so a burner fed its sea-level input at 1,000 m is putting the same fuel into less oxygen, and a buoyancy-vented appliance loses draft as well. CSA B149.1 deals with this in Clause 4.22, which Alberta Municipal Affairs reproduces in its STANDATA bulletin on high-altitude installations: for installations between 600 m and 1,350 m above sea level the appliance must be certified for high altitude to CSA 2.17 and adjusted to the high-altitude rating shown on its nameplate, following the manufacturer's instructions. What that adjustment is depends on the model: a smaller orifice, a lower manifold pressure on an appliance that carries its own regulator, a certified field conversion kit, or an air-supply adjustment on an induced-draft or direct-vent unit. The appliance is then marked with its altitude-adjusted input. The adjustment brings the input down, not up, so a larger orifice is the wrong direction and over-fires the burner. The appliance regulator cannot do the job on its own: it holds a gas pressure and has no way of sensing air density. And the utility sets its service pressure from its own distribution system, not from a customer's elevation. Above the highest elevation option the manufacturer provides, the same clause has the input reduced further following the manufacturer's instructions, and only where the manufacturer gives none does a fixed reduction of 4% for each additional 300 m apply. Prove the result with a combustion analysis and label the appliance with its adjusted rating.

Q18. Answer: B

Pre-purge exists to sweep the combustion chamber, the heat exchanger passages and the vent with fresh air before any ignition source is energized, so that any fuel that accumulated while the burner was off is diluted and carried out of the appliance. How long that takes is a volume problem: the fan delivers a known airflow at purge rate, the chamber and flue hold a known volume, and the purge must run long enough to deliver the number of air changes specified for that burner in the manufacturer's burner management sequence. A larger chamber, a longer flue, or a lower purge-rate airflow all lengthen the required pre-purge. The other three answers name real steps in a light-off sequence, but none of them is what the purge timer is sized from: ignitor warm-up, refractory warm-up and the flame-proving period are separate, independently timed events. Post-purge works the same way at the other end of the cycle, clearing residual products of combustion after shutdown.

Q19. Answer: D

Black steel pipe with malleable iron or steel fittings is the standard for above-ground gas piping inside a building: it is strong, it threads and welds readily, and it is the material the steel piping requirements of the installation code are written around. Galvanized steel is the same base product — ASTM A53/A53M is titled Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless, so black and galvanized pipe are two finishes of one specification — but galvanized is not the customary material for interior gas distribution and the zinc coating buys nothing in a dry indoor run. PVC is never a fuel gas material anywhere in the system: plastic gas piping in Canada is polyethylene to CSA B137.4, Polyethylene (PE) Piping Systems for Gas Services, and it is a buried outdoor material, not an indoor one. Copper tube is accepted for gas in specified applications and sizes, and it is common on short appliance runs, but it is not the usual choice for distribution piping through a building. Confirm the acceptable materials, the sizes and any restriction against the edition of CSA B149.1 in force in your jurisdiction together with the provincial amendments.

Q20. Answer: B

The inducer fan draws combustion gases through the heat exchanger and forces them out the vent pipe. It creates negative pressure in the heat exchanger, ensuring complete flue gas evacuation and allowing use of smaller-diameter vent pipes. A pressure switch verifies inducer is running before allowing ignition.

Q21. Answer: A

A low-water cut-off senses the water level in the boiler and interrupts fuel to the burner when the level falls below the safe minimum. Firing a boiler whose water level has dropped below the heated surfaces overheats the metal, which can crack the vessel or heat exchanger; on a steam boiler, feeding cold water back onto surfaces that have been fired dry can produce a violent release. The cut-off does not restore the water by itself unless it is paired with a feeder, so a low-water lockout is a signal to find the leak or the feed fault before resetting, not simply to reset and walk away. Each of the wrong answers belongs to a different control the same boiler may carry: loss of circulation is the flow switch's job, overpressure is the relief valve's job, and low return water temperature is handled by a boiler-protection or low-limit aquastat arrangement. Sorting these four apart is the point - they are all 'safety limits' but they sense different things.

Q22. Answer: B

Support spacing is not a single number to be carried in the head. The installation code sets it in a support-spacing table indexed by the size of the pipe, on the simple mechanical ground that larger, stiffer pipe spans further between hangers than small pipe does, so small-diameter pipe has to be supported more closely. Read the interval for the size and the material actually in front of you: tubing, plastic piping and vertical runs are each treated differently from horizontal rigid steel. Alberta's amendment to the support clause also permits spacing to ASME B31.1 or ASME B31.3 as alternatives to the table. Any answer that fixes one interval for every size and material is wrong for that reason alone, and the tight figure is as wrong as the loose one. Nor is pipe left to hang between fittings: under-supported pipe sags, and the bending stress lands in the threaded joints, which is where a gas line comes apart. Support does more than carry weight — it also holds the run against impact and vibration and keeps it in place if a joint is disturbed. On a rooftop the code adds a further requirement: piping less than NPS 1 must be supported horizontally every 4 ft (1.2 m), with vertical support to the table, and a support is provided at every threaded fitting.

Q23. Answer: C

Gas detector placement follows gas behaviour: Natural gas ($SG=0.55$, lighter than air) rises to the ceiling when leaked — detector within 300mm of ceiling. Propane ($SG=1.52$, heavier than air) settles to the floor and in low areas — detector within 300mm of floor. Wrong placement means the detector may not alarm until dangerous concentrations exist.

Q24. Answer: D

CSA B149.1 sets vent terminal height by two separate tests, and both have to be satisfied. First, the terminal must clear the point where the vent passes through the roof by the minimum the code gives for that roof's slope — the flatter the roof the smaller that distance, the steeper the roof the greater it becomes, which is why a flat roof has a low minimum rather than no minimum. Second, the terminal must rise above any portion of the building standing within the horizontal distance the code specifies — a parapet, a penthouse, an adjacent higher wall. A terminal that satisfies only the first test can still sit inside the wind eddy on the lee side of a taller structure, and that eddy is what drags flue gas back down into air intakes, windows and the building itself. The figure for a given job is read out of the code for the roof slope and the surrounding structures actually present, not assumed from a single remembered number, and the manufacturer's certified instructions may impose more.

Q25. Answer: D

A plastic vent is certified as a system, not as a shelf of parts. Saskatchewan's Codes of Practice for the natural gas and propane installation code puts it plainly at clause 8.10.4.1: 'ULC S636 PVC and CPVC glues, primers, pipe and fittings must be used only with the compatible companion products in order to be certified as a venting system. Inter-mixing of glues, primers, pipe or fittings from various manufacturers in any combination is prohibited.' Matching outside diameter does not make two products companion parts. The solvent cement is formulated for a particular resin and wall, and a joint made from mixed components sits outside the listing on which the vent's temperature and pressure rating depends. Tape does not repair that: the joint on a condensing or power-vented appliance is under positive pressure, and a wrap on the outside is not a bond. Neither does a passing tightness test, which shows the joint is tight today and says nothing about how it behaves hot, wet and acidic a year from now. Get the companion fitting for the system that is installed, and follow the vent manufacturer's certified instructions on compatibility.

Q26. Answer: B

Clause 6.3.2 of CSA B149.1, reproduced in full in Alberta Municipal Affairs' published comparison of the 2025 edition, requires a system supplied at up to and including 14 in WC to be designed so that the pressure loss between the appliance and either the termination of the utility installation or the last-stage regulator does not exceed the maximum allowable drop in Table 6.1, and it names the low-pressure capacity tables - A.1 and A.8 for a 0.5 in WC drop, A.2 and A.9 (or B.1 and B.6 for propane) for a 1 in WC drop - as tables which include allowance for a reasonable number of fittings. Two things follow. First, the length that governs is the run to the most remote outlet, because that is where the accumulated loss is greatest: measure from the point of delivery - the meter outlet or the last-stage regulator - to the farthest appliance, find that length in the table or the next larger length listed, and use that one row for every section of the system, sizing each section for the total load it carries. British Columbia's trades training text sets the procedure out the same way: calculate the longest measured run from the meter to the most distant appliance, select the table row equal to or greater than it, and size each pipe from that row. Sizing each section on its own short length ignores the loss upstream of it and starves the far appliance when several run together; adding every section's length together overstates the loss and oversizes the work. Second, fittings are not added when these tables are used. Their allowance is already built into the table capacities, and the same training text says a separate fitting allowance calculation is not required for them. Converting elbows, tees and valves to equivalent length of straight pipe belongs to the Annex calculation method and to systems above 14 in WC, where Clause 6.3.3 tells the designer to make allowance for fittings as necessary. The allowable pressure drop itself is read from Table 6.1 against the system supply pressure, and some provinces tighten it in their codes of practice, so confirm the figure for the jurisdiction before choosing the table.

Q27. Answer: A

Proper commissioning sequence: 1) Verify supply gas pressure, 2) Inspect and purge new piping, 3) Light appliance per manufacturer instructions, 4) Verify manifold/burner pressure, 5) Verify all safety controls (high limit, pressure switches), 6) Conduct combustion analysis (CO₂, O₂, CO, flue temp), 7) Document all readings. Always follow manufacturer commissioning checklist.

Q28. Answer: C

Standard white PTFE tape is not rated for fuel gas service. CSA B149.1 Clause 6.9.6 requires that when a jointing sealant is used it shall be certified to CAN/ULC-S642 and shall be applied to the male threads of a metal pipe, and that tape shall be stretched and applied in a clockwise direction with a 50% overlap leaving the first two starter threads bare. Gas-rated PTFE tape is normally yellow, and gas-rated pipe thread compounds carry the same CAN/ULC-S642 certification; a general-purpose sealant chosen off the shelf has no such certification and can soften or wash out in the presence of gas and its liquids, giving a joint that passes the pressure test and weeps months later. Leaving the first two threads bare keeps sealant from being carried into the bore where it can foul a control or an orifice. Tapered NPT threads do not seal on their own either: the thread form leaves a spiral leak path, and filling it is exactly what the sealant is for.

Q29. Answer: D

The two regulators sit in series and each takes the pressure down one step. The service regulator at the meter set reduces the utility's distribution pressure to the service pressure delivered into the building, about 7 in WC (1.75 kPa) for natural gas. By the time the gas reaches an appliance it has lost a little more in the building piping under flow, so the appliance inlet pressure is nominally about 7 in WC and has to sit within the minimum and maximum marked on the rating plate. The appliance regulator, built into the combination gas valve or fitted on the supply, then takes that inlet pressure down to the manifold (burner) pressure the manufacturer specifies: about 3.5 in WC for natural gas and about 10 in WC for propane. Inlet and manifold pressure are measured at two different points and are never the same number, because a regulator can only reduce a pressure that is already there. So the appliance regulator handles the lower of the two pressures, not the higher; the service regulator sits at the meter set, normally outdoors, and the appliance regulator sits at or inside the appliance, so the indoor-outdoor claim is backwards; and the two are not interchangeable, since each is sized and sprung for its own inlet and outlet range. On propane the second-stage regulator at the building plays the service role, delivering about 11 in WC (2.74 kPa) to the appliance inlet.

Q30. Answer: B

The Red Seal trade descriptions draw the line at appliance and equipment input rating. Gasfitters Class A work on appliances and equipment including those exceeding 400,000 Btuh (120 kW), which brings in boilers, burners, makeup air units, process burners and similar large equipment; Gasfitters Class B work on fuel-fired appliances that do not exceed that rating, in residential, institutional and commercial buildings and restaurants. The two are separate certifications rather than apprentice and journeyman stages of one ladder, neither is defined by a single fuel since both work with the fuel gases their jurisdiction licenses them for, and gas fitting is a regulated trade across Canada in either class. Licence titles, classes and the exact scope each one carries are set by the provincial or territorial gas authority, so confirm local scope before working outside the class you hold.

Q31. Answer: A

A steady blue flame indicates complete combustion with the correct air-fuel mixture. Blue comes from excited CH radicals. Yellow/orange flames indicate insufficient air or incorrect mixture (incomplete combustion) — carbon particles glow yellow before they can fully oxidize, producing soot and CO. A few yellow tips at the top of flame is acceptable; fully yellow flame is not.

Q32. Answer: A

Maximum theoretical CO₂ for natural gas (stoichiometric) is approximately 11.7-12.5%. In practice, residential furnaces read 8-10% CO₂ due to excess air, which dilutes the CO₂ concentration and increases flue gas volume and heat loss. A reading of 8.5% indicates significant excess air (approximately 35-40% excess). Higher excess air = more heat wasted heating unused air. Target: 8.5-10% CO₂ for natural gas furnaces.

Q33. Answer: A

Clause 6.16.4 of CSA B149.1 sets the trigger by distance, not by material, pressure or length: 'Tubing run inside hollow walls or partitions within 1.75 in (43 mm) of the surface shall be protected against physical damage and puncture at the joists, studs, and plates by the use of No. 16 USG (1.59 mm) plates or sleeves.' The logic is the fastener. A drywall screw or trim nail driven at a stud or plate reaches only so far, so tubing lying deeper than 43 mm from the surface is out of its path, while tubing lying closer must have steel between it and the fastener. The clause adds that it does not apply to tubing that passes directly through a wall or partition. It applies to tubing of every kind the code recognizes, which Clause 6.2.8 lists as CSST, seamless copper and seamless steel, so copper gets no exemption on account of its wall, and neither the operating pressure nor the length of the run enters into it; the 6 ft (2 m) figure a candidate may recall is the support spacing for tubing in Table 6.2, not a protection trigger. For CSST there is a second layer. Clause 6.16.13 (2015 numbering) requires CSST and its fittings to be protected against physical damage in accordance with the manufacturer's certified installation instructions and with the code, and Alberta's gas safety bulletin G-04-01 repeats that CSST 'shall be protected against physical damage in accordance with the manufacturer's certified installation instructions'. Those instructions can protect a wider zone than the code minimum. Manitoba's gas equipment bulletin ITSM 17-011 states the same 16-gauge plate rule for tubing less than 1 3/4 in from the exposed edge of a stud, joist or plate, and then says that where the manufacturer's installation instructions exceed that minimum, the manufacturer's instructions are to be followed. Concealed tubing must also be inspected and pressure tested in its final position before the wall is closed (Clauses 6.7.1 and 6.22.5).

Q34. Answer: A

Clocking the meter is a volume measurement and nothing else. With every other gas appliance off, you time how long the smallest test dial on the meter takes to complete one revolution; that dial is marked in a known volume, so the timing gives cubic feet per hour — the volume of gas the appliance is actually drawing. Input is then a calculation rather than a reading: multiply the measured flow by the heating value of the gas to get BTU per hour, and compare that against the input on the rating plate. Nothing in the test reads pressure. Manifold pressure is read with a manometer at the burner test port and inlet pressure at the appliance inlet port, which is a separate check. Nothing in it reads heating value either — that is a property of the gas the utility supplies, and it has to be obtained from the utility if the result is to be accurate. And efficiency is not measured here at all: clocking says how much fuel goes in, not how much of it becomes useful heat, which takes a combustion analysis.

Q35. Answer: D

Natural gas explosive range: LEL = 5% (minimum concentration at which gas will ignite if an ignition source is present) to UEL = 15% (maximum ignition concentration). Below 5% = too lean to ignite. Above 15% = too rich to ignite (but becomes explosive as it disperses). Gas detectors typically alarm at 10-25% of LEL (~0.5-1.25% actual concentration) to provide warning well before reaching the explosive range.

Q36. Answer: B

As gas flows through a pipe, friction causes pressure to drop, and the greater the flow rate, the greater the pressure drop per unit length. Pipe sizing ensures that pressure drop from the supply (meter) to the most distant appliance stays within allowable limits (125 Pa for residential NG), maintaining adequate appliance inlet pressure for proper combustion.

Q37. Answer: A

The standard low-pressure natural gas service to a house is about 7 inches of water column (7 in WC, 1.75 kPa). That is the outlet setting of the service regulator at the meter set, and it is the pressure delivered into the building piping. It is not the same number at every point downstream. At the appliance inlet the pressure is nominally about 7 in WC less whatever the building piping drops while gas is flowing, which is why inlet pressure is read at the appliance with the appliance firing, and why the rating plate states a minimum and a maximum inlet pressure rather than a single figure. The appliance's own regulator, inside the gas valve, then reduces the inlet pressure to the manifold (burner) pressure, about 3.5 in WC (0.87 kPa) for natural gas, so 3.5 in WC is a burner figure, not a service figure. 14 in WC is twice the service setting; it may appear as a lockup reading at zero flow, but it is not the delivered service pressure. 2 psig (13.8 kPa) is an elevated-pressure service that a utility may offer for larger loads, with a regulator at the building bringing it back down.

Q38. Answer: B

Power-vent (Category III) appliances use a blower to push flue gases through the vent. Maximum horizontal run depends on the appliance manufacturer's specifications and the vent system's static pressure capacity — typically up to 15-25 m depending on the number of elbows and pipe size. Each elbow reduces equivalent straight-pipe capacity. Exceeding maximum run reduces flue gas velocity and can cause condensate accumulation.

Q39. Answer: D

To check meter adequacy: convert BTU/h to ft³/h by dividing by 1,000 (BTU/ft³). 400,000 BTU/h ÷ 1,000 = 400 ft³/h, which exceeds the meter capacity of 250 ft³/h — the meter is insufficient. A larger meter (e.g., 400 ft³/h or larger) is required. Note: diversity factor may apply for large installations.

Q40. Answer: C

Natural gas combustion produces CO₂ and H₂O. The dew point of natural gas flue gases is approximately 55-60°C (130-140°F); below this temperature, water vapour condenses into liquid in the vent. The condensate absorbs CO₂ forming carbonic acid (H₂CO₃) and sulfurous acid, producing an acidic liquid (pH 3-4). This is why ordinary metal vents corrode when serving Category IV condensing appliances.

Q41. Answer: D

Purging replaces air in new gas piping with gas by allowing gas to slowly displace air at a low-pressure vent point or appliance connection. CSA B149.1 requires purging using a controlled vent to an unconfined outdoor location — never into an enclosed space (explosion risk). Monitor with a combustible gas detector until gas composition is confirmed. The first gas/air mixture vented may be in the flammable range — keep ignition sources away.

Q42. Answer: C

Stoichiometric combustion uses exactly the theoretical air-fuel ratio to completely combust all fuel — no excess air and no incomplete combustion: all fuel molecules react with exactly enough oxygen molecules, producing only CO₂ and H₂O with no excess O₂ remaining. It is a calculation, not a setting anyone runs a burner at. In practice gas appliances are set up with roughly 15% to 50% excess air (about 3% to 7.5% O₂ in the dry flue gas), power burners at the low end and residential atmospheric appliances at the high end, because mixing is never perfect and any shortfall of air makes carbon monoxide.

Q43. Answer: A

Condensing furnaces extract so much heat that flue gases cool below the dew point, producing liquid condensate. The condensate contains carbonic acid ($\text{CO}_2 + \text{H}_2\text{O}$) and sulfurous acid, making it acidic (pH 3-4). Many municipalities require a condensate neutralizer before discharging to certain drains, such as copper or cast iron drain systems.

Q44. Answer: D

HSI (hot surface ignitor) is made of silicon carbide or silicon nitride. The control board energizes it for a warm-up period (15-30 seconds) until it glows red-hot, then opens the gas valve. The glowing element ignites the gas. HSI replaced standing pilots in mid-efficiency and high-efficiency furnaces.

Q45. Answer: A

A drip leg is a short vertical pipe extension below the horizontal run, capped at the bottom, installed as the last fitting before connecting to the appliance. Condensate, pipe scale, and debris fall into the drip leg before they can enter the appliance regulator and gas valve, preventing blockage and damage.

Q46. Answer: C

LOTO for gas systems prevents accidental re-energization of the gas supply while work is in progress: 1) Close the supply valve, 2) Lock the valve in the closed position with a lockout device, 3) Attach a danger tag with worker's name and warning, 4) Verify zero energy (test downstream for gas flow/pressure). This prevents another worker from inadvertently opening the gas supply during maintenance.

Q47. Answer: C

Common venting of two Category I appliances is permitted under specific conditions. The common vent segment must be sized using CSA B149.1 common vent sizing tables for the combined total BTU input, considering: height of the vertical common vent, individual connector lengths, and both appliance types. Improper sizing causes spillage at one appliance.

Q48. Answer: D

A listed vent cap sits on the terminal to keep rain, snow and debris out of the vent and to shield the outlet from wind, so that gusts do not push down the flue and interrupt draft. Water standing in a vent corrodes it and can drain back into the appliance, and a wind-induced downdraft can stall the flue and spill combustion products into the room. A cap does not clean the flue gas: nothing in a vent removes carbon monoxide, which is why CO is controlled by correct combustion and a sound vent rather than by hardware at the top. Caps are not interchangeable either — use the cap listed with that vent system, because a restrictive or unlisted cap adds resistance and reduces draft.

Q49. Answer: C

Flame rectification uses the ionizing property of combustion gases. An AC voltage is applied between the flame rod (sensor) and the burner ground. The flame acts as a diode — it rectifies AC to DC, so DC current flows only when flame is present, proving burner ignition. The control board detects the DC current (microamps); if flame fails, DC stops, and the board shuts the gas valve within 1-2 seconds.

Q50. Answer: D

Gas dryer exhaust carries moisture, heat, lint, and combustion products including carbon monoxide. The building code article on venting of laundry-drying equipment requires the exhaust duct or vent connected to that equipment to discharge directly to the outdoors, and requires the duct to be independent of other exhaust ducts, accessible for cleaning, and made of a smooth corrosion-resistant material. Discharging into an attic, crawl space, garage, wall cavity, or any lint trap inside the building creates a lint fire hazard, moisture damage, and a carbon monoxide risk.

Scored under 70%?

The real exam draws from a much wider pool. The **Gasfitter (Class A) Complete Question Bank** gives you every question on our platform with full explanations and key-concept summaries — built to close exactly the gaps this mock exposed.

Get the Gasfitter (Class A) Complete Question Bank →

Or keep practising free online at redsealquiz.ca/gasfitter-class-a.html

© 2026 · redsealquiz.ca/gasfitter-class-a.html · For personal study use only. Independent, unofficial study resource — not affiliated with or endorsed by the CCDA, the Red Seal Program, or any apprenticeship authority. See "About This Material".